

Aditya Kulkarni

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PROFESSIONAL SUMMARY

Backend engineer with hands-on Kafka streaming infrastructure across three production systems -- a Kafka-driven corpus indexing pipeline (VectorLift), a Kafka+Temporal fault-tolerant async worker at 2,000 inference requests/second (AlgoPulse), and Kafka streaming for real-time ticket ingestion (Capgemini). Built distributed data pipelines from scratch and delivered 394 commits and 7 production deployments at Isomer AI in a fast-paced startup environment. Python and TypeScript across the full stack; comfortable owning streaming systems end-to-end from design through production.

CORE COMPETENCIES

Apache Kafka

Distributed Streaming Systems

High-Throughput Data Pipelines

Python + FastAPI

TypeScript + Node.js

Temporal.io + Fault Tolerance

PostgreSQL + Redis

Prometheus + Grafana

WORK EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern - Raleigh, NC

- Owned end-to-end delivery of 10+ backend features across **394 commits and 7 production deployments**; built real-time event pipelines (Pusher), Temporal.io workflow orchestration, and TypeScript + Node.js + PostgreSQL + Redis data layer
- Designed composable REST APIs with Auth0 RBAC, Docker containerization, GitHub Actions CI/CD, and Sentry production observability -- operated with high autonomy in a fast-paced startup environment

NC State University Apr 2025 - Present

Graduate Research Assistant - Raleigh, NC

- Built a GCP Cloud Run platform for 100+ researchers to explore 500GB genomic datasets; improved ML model accuracy by 12% through hyperparameter tuning

Capgemini Jun 2023 - Aug 2023

AI/ML Intern

- Implemented **Kafka streaming + Spark batch + Hadoop** distributed data pipelines for real-time support ticket ingestion; reduced latency by 20% and processing time by 15%

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern

- Built 5+ AI-driven automation workflows; automated AWS and GCP deployments; integrated Grafana monitoring dashboards

PROJECTS

VectorLift -- Kafka-Driven Semantic Search Pipeline

Kafka + FastAPI + Python

github.com/Aditya-k24/VectorLift

Built a Kafka-driven indexing pipeline for event-based corpus ingestion into Elasticsearch and Qdrant vector store -- handles document arrival events, routes to appropriate indexing workers, and maintains consistency across stores. FastAPI search server with per-stage latency tracking; Prometheus + Grafana dashboards for throughput, lag, and error rates. Fine-tuned sentence-transformers with in-batch negatives on MS MARCO; 6 pipeline configurations evaluated with NDCG@10, MRR@10, MAP, and bootstrap significance testing.

Python, Kafka, FastAPI, Elasticsearch, Qdrant, PyTorch, Sentence-Transformers, PostgreSQL, Redis, Prometheus, Grafana, Docker

AlgoPulse -- High-Throughput Async Worker System

Kafka + Temporal + 2K req/sec

github.com/Aditya-k24/AlgoPulse

Event-driven worker system with Temporal.io and Kafka for state synchronization and fault tolerance at **2,000 async inference requests/second** without state loss. Identified API-polling bottleneck under load and re-architected to Supabase Edge Functions for streaming delivery -- same diagnosis-and-redesign pattern needed for streaming systems at scale.

TypeScript, Node.js, Supabase Edge Functions, Temporal.io, Kafka

PulseRank -- Real-Time Event-Driven Platform

TypeScript + NestJS + SSE

github.com/Aditya-k24/PulseRank

Event-driven leaderboard and activity feed: MySQL transactional writes with idempotency-key deduplication and outbox pattern, MongoDB feed projections with TTL expiry, Redis sorted sets for sub-millisecond reads, live updates over Server-Sent Events. Multi-store consistency verified with Testcontainers; OpenTelemetry + Prometheus observability; k6 load benchmarking.

TypeScript, NestJS, Next.js, MySQL, MongoDB, Redis, Docker, OpenTelemetry, Prometheus, k6, GitHub Actions

Inventory Reservation Service -- Concurrency-Safe API

Java 21 + Spring Boot

github.com/Aditya-k24/InventoryReservationService

Transactional inventory API with PostgreSQL row-level locking (SELECT FOR UPDATE) preventing overselling across concurrent reservations; verified with Testcontainers integration tests. Idempotent endpoints, Flyway migrations, Spring Security RBAC, GitHub Actions CI/CD.

Java 21, Spring Boot 3.5, PostgreSQL, Flyway, Spring Security, Testcontainers, Docker, GitHub Actions

EDUCATION

North Carolina State University -- MS Computer Science

Aug 2024 - May 2026

Raleigh, NC | Courses: Artificial Intelligence, Operating Systems, Software Engineering, Algorithms, Data Science

Mumbai University -- BTech Computer Science

Aug 2020 - May 2024

CGPA: 9.32/10 | Courses: Data Structures & Algorithms, Machine Learning, Software Engineering, Big Data Infrastructure

SKILLS

Streaming: Apache Kafka, Temporal.io, event-driven architecture, high-throughput pipelines, fault tolerance

Languages: Python, TypeScript, Java, JavaScript, SQL

Backend: FastAPI, Node.js, NestJS, Spring Boot, REST APIs, PostgreSQL, MySQL, MongoDB, Redis, Docker

Observability: Prometheus, Grafana, OpenTelemetry, Sentry, k6 benchmarking

Cloud: AWS, GCP (Cloud Run), GitHub Actions CI/CD, Kubernetes, Spark, Hadoop